

NANDU PANAKANTI

Software Development Engineer | Distributed Systems · Cloud-Native APIs · Java · AWS

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[GitHub](#) · [LinkedIn](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

Full-stack Software Engineer with production ownership of distributed microservices, event-driven systems, and cloud-native APIs. Delivered measurable impact including 25% latency reduction in robotics telemetry APIs, 40% notification latency improvement using Redis caching, and complete automation of billing operations in a multi-tenant SaaS platform. Strong foundation in backend engineering, distributed systems, cloud infrastructure, and AI-assisted development workflows. Experienced in designing scalable systems, optimizing APIs, and deploying production-grade applications.

TECHNICAL SKILLS

Languages:	Java, Python, JavaScript, SQL, PHP
Backend:	Spring Boot, Node.js, Express.js, REST APIs, Microservices, FastAPI
Frontend:	React.js, Bootstrap, jQuery, AJAX, HTML5, CSS3
Distributed Systems:	Apache Kafka, Redis, Event-Driven Architecture, Dead-Letter Queues, Idempotent APIs
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda, CloudFront), Docker, GitHub Actions, CI/CD, Render
Databases:	PostgreSQL, MySQL, MongoDB, MongoDB Atlas, Schema Design, Query Optimization
AI / GenAI:	Claude API, GitHub Copilot, Prompt Engineering, SHAP Explainability, ML Pipeline Integration
Security:	JWT Authentication, RBAC, bcrypt, Stripe Webhooks, Prepared Statements
ML & Data:	Scikit-learn, Random Forest, Pandas, NumPy, Streamlit, SHAP, SMOTE
System Design:	Kafka, Redis Caching, DB Sharding, Consumer Groups, Circuit Breakers, Rate Limiting

EXPERIENCE

Backend Developer Intern — Jio Robotics

Sep 2025 – Jan 2026

Java · Node.js · Express.js · REST APIs · JWT · Performance Optimization

- Took full ownership of 10+ production REST APIs serving real-time robotics telemetry systems, improving average response latency by 25%.
- Developed scalable backend services using Java and Node.js following modular and clean architecture principles.
- Improved system reliability by implementing async error handling, validation layers, and unit testing.
- Hardened API security using JWT authentication and role-based authorization.
- Collaborated in Agile sprint cycles and contributed to production debugging and deployment.

PROJECTS

LeaseHub — Multi-Tenant Property Management SaaS

Oct 2025 – Jan 2026

Node.js · React.js · MongoDB · Stripe · JWT/RBAC · Docker · CI/CD

[GitHub](#) · [Admin Demo](#) · [Tenant Demo](#)

- Architected and deployed a production-oriented multi-tenant SaaS platform automating lease workflows.
- Implemented automated billing pipelines eliminating 100% manual operations.
- Integrated Stripe Payments with webhook verification and idempotent processing.
- Designed RBAC authentication using JWT and bcrypt.
- Optimized MongoDB schema improving query performance by 30%.
- Deployed using Docker, CI/CD, Render, and MongoDB Atlas.

Employee Attrition Prediction System

Oct 2025 – Nov 2025

Python · Scikit-learn · Random Forest · SHAP · Streamlit · Pandas

[GitHub](#) · [Live Demo](#)

- Built an ML application to predict employee attrition using Random Forest.

- Developed preprocessing pipelines including SMOTE for imbalance handling.
- Integrated SHAP explainability dashboards.
- Deployed on Streamlit Cloud.

AI Workflow Automation Platform

Apr 2026 – May 2026

FastAPI · PostgreSQL · React · Prometheus

[GitHub](#) · [Live Demo](#)

- Designed and built an AI-driven workflow system using asynchronous task pipelines and background workers.
- Implemented retry logic, exponential backoff, and Dead Letter Queue (DLQ).
- Developed AI-powered log analysis and recommendation system.
- Engineered an AI evaluation layer to score and validate outputs.
- Built React dashboard for real-time monitoring and visualization.
- Integrated observability metrics (latency, retries, failures).
- Deployed full-stack system on Render and Vercel.

StudyMate — Learning Collaboration Platform

Sep 2025 – Dec 2025

PHP · MySQL · Bootstrap · jQuery · AJAX

- Built a peer-learning collaboration platform with scheduling and admin workflows.
- Designed normalized MySQL schemas with optimized joins.
- Implemented secure authentication using bcrypt and prepared statements.
- Added AJAX-based real-time validation.

Distributed Notification Service

Feb 2026 – Apr 2026

Java · Spring Boot · Apache Kafka · Redis · Docker

- Built distributed async notification system using Kafka.
- Implemented Redis-based deduplication and DLQ handling.
- Reduced latency by 40% using caching.
- Designed modular microservices architecture.

AWS IAM Drift Detection System

Jan 2026 – Feb 2026

AWS Lambda · CloudTrail · EventBridge · SNS · DynamoDB

[GitHub](#)

- Built real-time IAM security monitoring system.
- Implemented event-driven workflows using AWS services.
- Designed alerting using SNS.
- Applied least-privilege IAM design.

EDUCATION

M.S. Computer Science — University of Central Missouri, Warrensburg, MO

May 2026

Relevant Coursework: Distributed Systems, Cloud Computing, Database Systems, Algorithms, Operating Systems

CERTIFICATIONS

- AWS Cloud Architecting [\[View\]](#)
- AWS Cloud Security Foundations [\[View\]](#)
- Claude API Development [\[View\]](#)
- Walmart USA — Advanced Software Engineering Job Simulation [\[View\]](#)
- Skyscanner — Software Engineering Job Simulation [\[View\]](#)
- NPTEL DBMS, Python, Embedded Systems [\[View\]](#)

ADDITIONAL INFORMATION

- Solved 200+ LeetCode problems (Arrays, Sliding Window, Trees, Graphs, DP, Binary Search, Heaps).
- Strong understanding of distributed systems concepts.
- Experience with CI/CD, debugging production systems, and Agile workflows.
- Active user of AI-assisted development tools (Claude API, GitHub Copilot, ChatGPT).